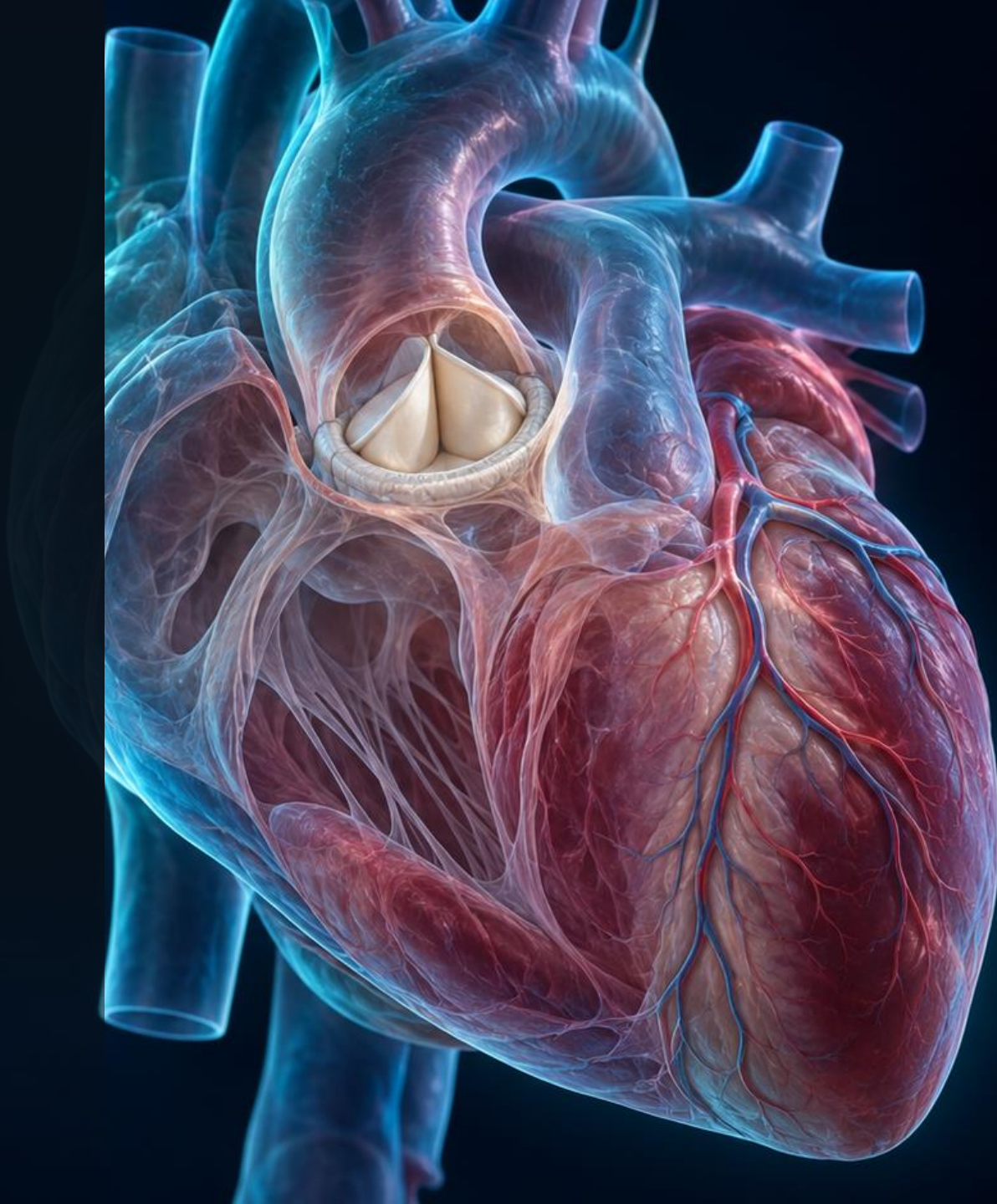

VALVE-WATCH

Risk-adapted surveillance
after bioprosthetic AVR

TEAM POLYPHYMOS

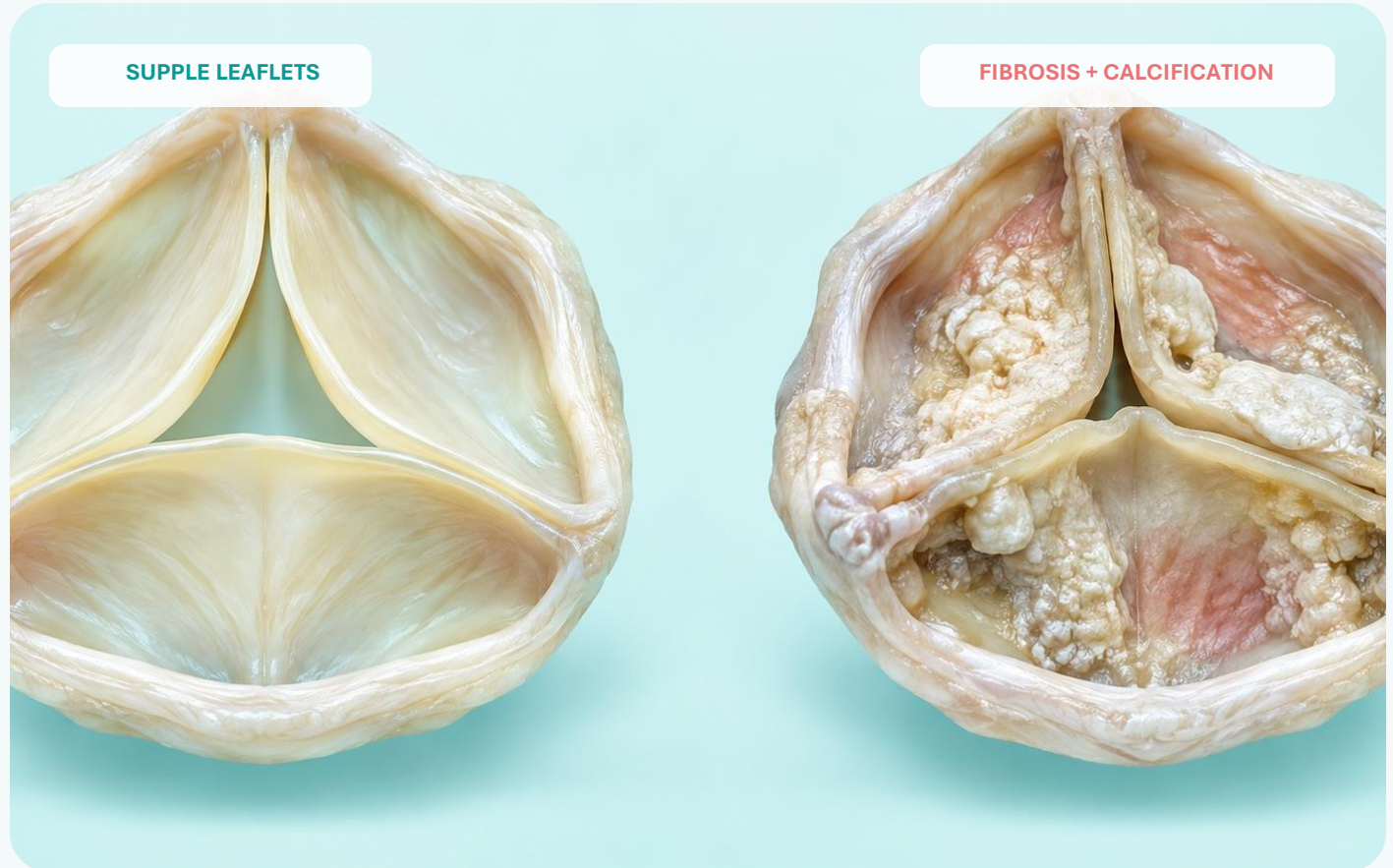


Bioprosthetic valves fail along different timelines

Structural valve deterioration changes the valve itself

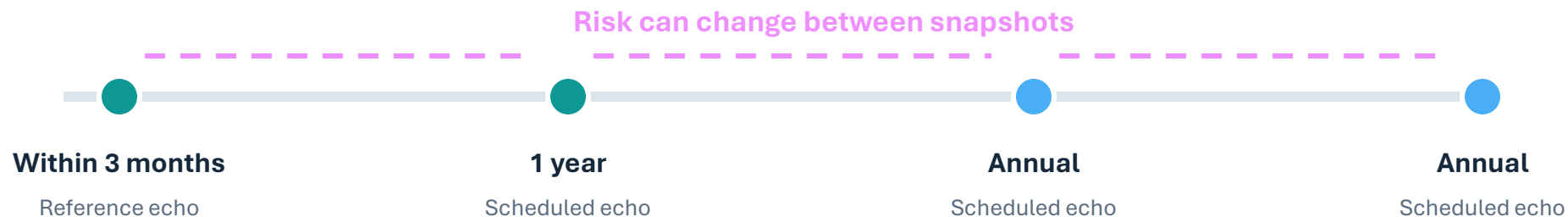
Leaflets may thicken, calcify, tear, or lose motion. The result can be stenosis, intraprosthetic regurgitation, or both.

Implant age and valve condition often diverge.



Guideline surveillance still leaves a care gap

Guideline baseline



66%

discharged after the six-week review with no planned hospital follow-up

Single UK centre; locally eligible surgical-valve cohort from 2008

The schedule is clear. The record often is not.

Missed or external studies

Symptoms documented between visits

Echo measurements spread across systems

Late recognition of a changing trajectory

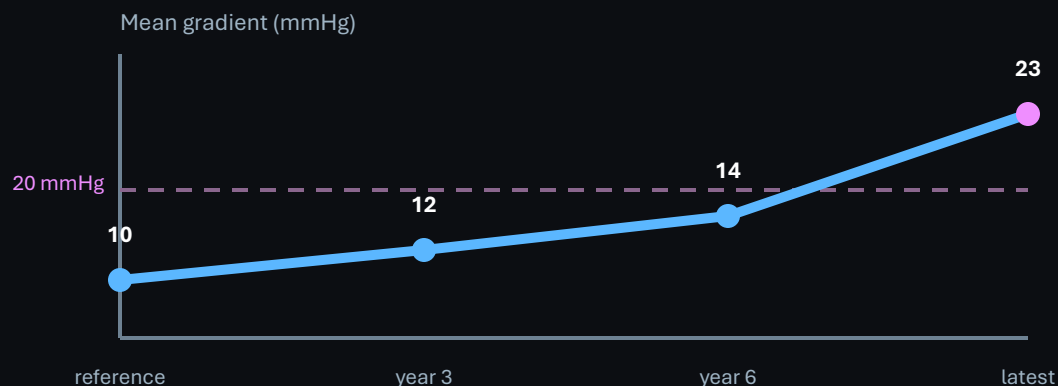
2025 ESC/EACTS guideline; Alaour et al., 2018

Two different reasons for earlier review

CURRENT DYSFUNCTION

HVD detected; SVD not yet confirmed

76 y • SAVR • 8.1 years since implant • new exertional dyspnoea



+13 mmHg and DVI fell 0.41 to 0.30

VARC-3 stage 2 HVD criteria met. Structural mechanism not documented in the available record.

NEXT: verify source echo, assess thrombosis or endocarditis, obtain morphology

No future-risk score while the current event remains unresolved

FUTURE RISK

No HVD today; elevated 3-year risk

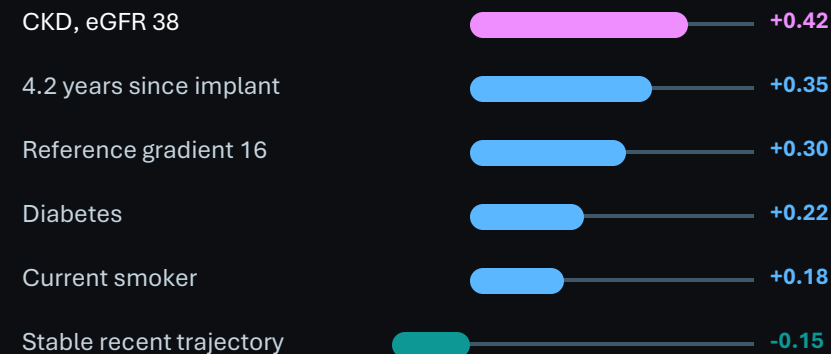
63 y • SAVR • 4.2 years since implant • stable current echo

17% illustrative conditional 3-year SVD risk

HIGH PRIORITY

SHAP-STYLE EXPLANATION

Illustrative contributions to model log-odds



Known early HVD correlates include diabetes, active smoking, renal insufficiency and a reference gradient ≥ 15 mmHg.

NEXT: clinician review and earlier echo consideration. This output does not diagnose SVD.

Study population and endpoint hierarchy

PRIMARY COHORT

Adults with confirmed bioprosthetic SAVR or TAVR

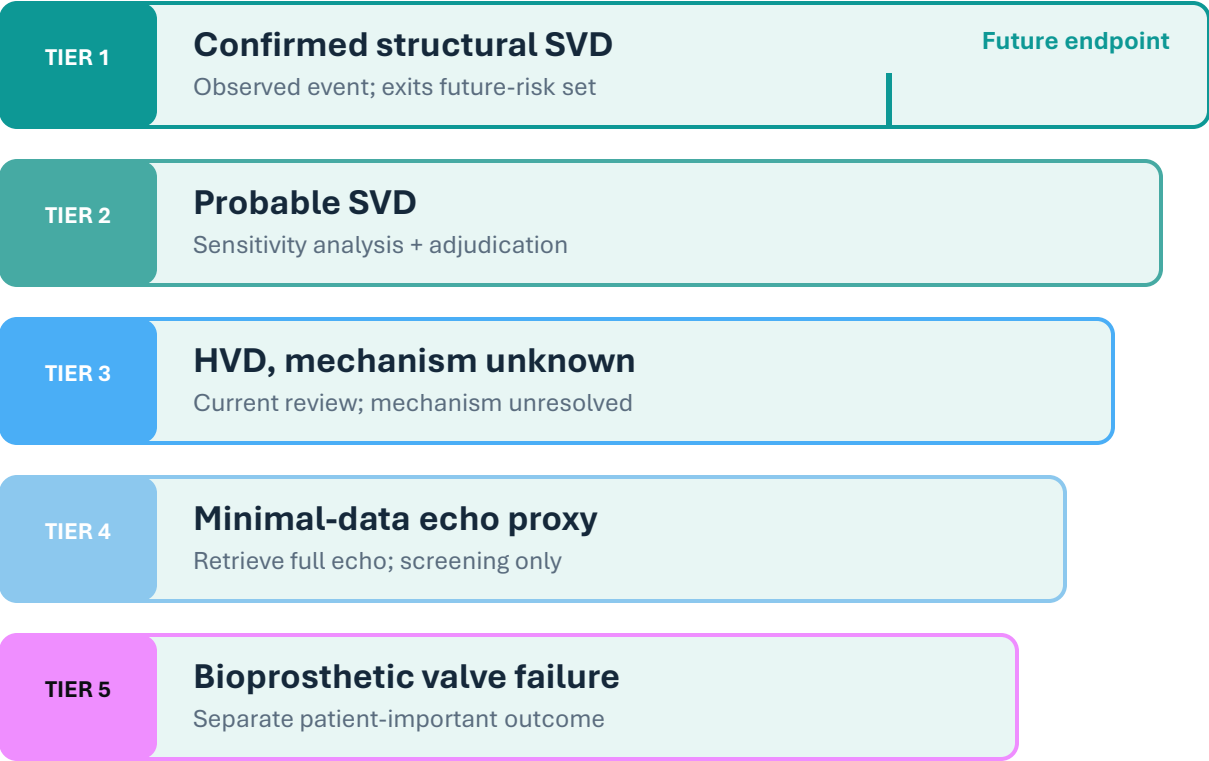
Stable reference echo at 30–90 days

Flagged fallback within 180 days

Primary exclusions

- Mechanical aortic valve
- Index redo or valve-in-valve procedure
- Severe alternative dysfunction at baseline

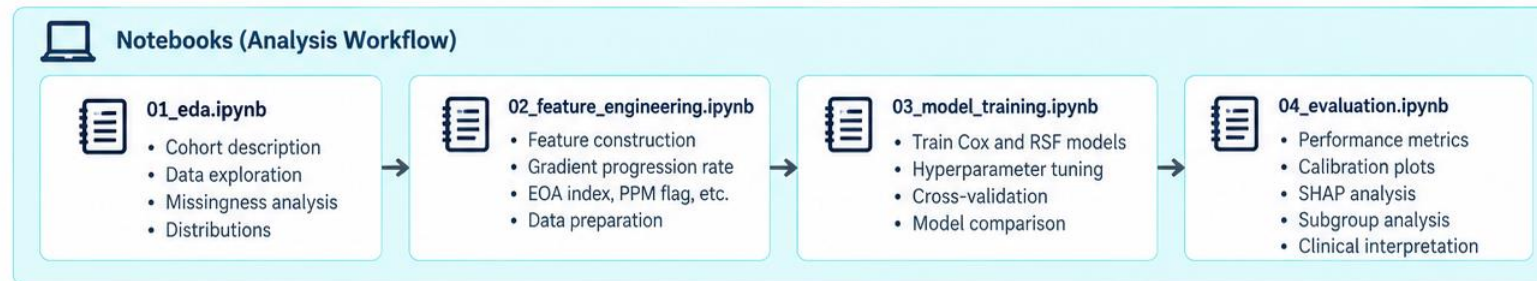
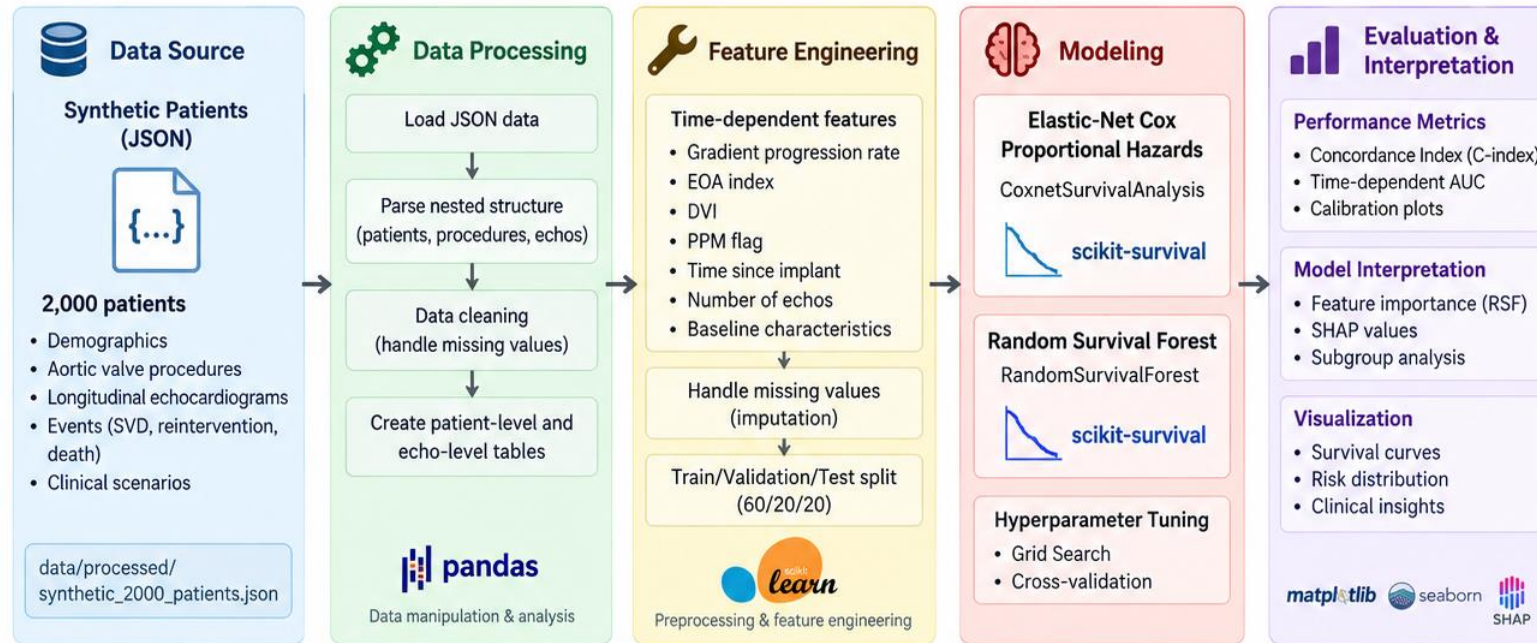
VARC-3 EVIDENCE LADDER



VALVE-WATCH

AI for Long-Term Aortic Valve Durability Surveillance Project Architecture

From longitudinal echo data
to personalized risk prediction



From fragmented records to a longitudinal valve story



ONE PASS THROUGH EACH SOURCE

MedGemma extracts explicit, source-linked facts. It does not classify endpoints or predict risk.

Laboratory and medication rows enter directly with dates, values, units, and source IDs.

Deterministic code calculates PPM, endpoint tiers, current-status routing, censoring, and landmark eligibility.

117

patients

215

notes

93

echo-screened

43,550

lab rows

5,807

medication rows

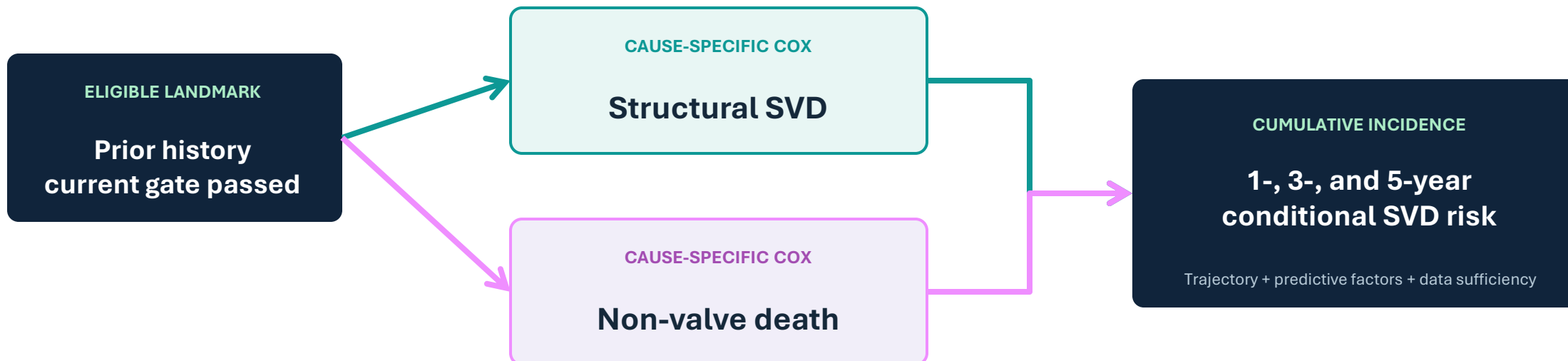
Labs + medications cover only 17 patients

Rolling-landmark competing-risk prediction

Only resolved, event-free landmarks enter the model

ENGINEERING TEST DATA

2,000 synthetic patients • 13,010 landmarks



COMPARATORS

Random survival forest

Simple Cox: implant age + approach + valve size + baseline gradient

Same-visit event evidence never becomes a predictor. Synthetic data test the code, not clinical performance.

Validation around events and clinical decisions

300

confirmed SVD events
for model development

≈ 4,500–7,000 patients

100

event floor for independent
validation

Final size set by precision

CLINICAL VALIDATION DATASET

Multi-site AVR cohort

Linked longitudinal echo, implant, outcome and mortality
records

Earlier years/sites develop • later period/site tests

Performance

Uno's C-index time-dependent AUROC integrated Brier score calibration decision-curve benefit

PROJECT-SPECIFIC GO / NO-GO

≥ 0.70

C-index

0.8–1.2

calibration slope

$\geq 85\%$

3-year sensitivity

$\leq 25\%$

patients flagged

Study criteria, not universal clinical standards

VALVE-WATCH

Earlier signal **A changing gradient, new regurgitation or high-risk trajectory enters a clinician review queue.**

Planned response The valve clinic can confirm the mechanism and bring imaging or specialist review forward.

The goal

Aim to preserve function and improve quality of life by shifting care toward planned follow-up.

Potential system benefit: more efficient surveillance, with fewer avoidable emergency presentations and urgent procedures.

